COLDSTART: Anticipating and quantifying health-system crises before the outcome is observable

1. Summary

When a health crisis begins, the quantities decision-makers need to forecast (calls, presentations, demand, occupancy, severity) are precisely those for which almost no local outcome data yet exist: the outcome series is a handful of noisy observations, and systems that perform well on years of local history are weakest when early decisions matter most.

There is another source of quantitative information: published evidence from analogous events elsewhere. It is rarely tested as formal prior information, because of the lack of reliable extraction and the difficulty of transporting it to the new setting. A confidently wrong prior is worse than none: it is most influential exactly when local data cannot correct it.

This project asks one end-to-end question: **can published quantitative evidence be reliably extracted, transported and used to forecast health-system escalation when local outcome data are insufficient at crisis onset; can we detect early when it should not be trusted; does any gain change operational decisions; and can the whole loop run automatically, prospectively verified, under pre-registered rules? LiteRev-Evidence**, the prototype I built, already walks this pipeline from a plain-language question to certainty-graded extraction, pooled parameter distributions and tuned candidate models. The project turns it into a validated instrument and tests it. I will (1) establish the error structure of automated extraction and propagate it into evidence-derived priors, for parameter values and the rest of what the literature carries (explanatory variables and lags, model forms, outcomes and thresholds); (2) represent the state of the care system as an unobserved regime process whose escalation probabilities carry every input's uncertainty, fed by those priors and a critical-slowing-down indicator read from the short local history; (3) test, by strict rolling-origin reconstruction of each historical moment's information, whether the evidence improves cold-start forecast skill and when it causes harm; and (4) whether any gain is large enough to change operational decisions. The decisive experiment runs a pre-registered library of candidate models (local baselines, surveillance algorithms, machine-learning and regime models; mechanistic components only for epidemiological crises) through an automated engine that checks assumptions, scores on established metrics and selects each arm's champion under rules fixed in advance. One contrast defines the confirmatory family: the evidence-informed against the local-only champion, tested on respiratory episodes and, only if that passes, repeated on heat as the sequential generalisation.

The forecasting machinery is validated first on open Swiss series (a public benchmark), then on Geneva's operational series. Three Geneva crisis archetypes, and no more, carry the real-world tests: respiratory epidemics confirmatory, heatwaves the sequential generalisation, legionellosis a scoped year-4 extension. The outcome is intentionally falsifiable: if the evidence improves cold-start forecasts, the project establishes when and by how much; if not, or if harm cannot be detected early enough, a boundary condition for automated evidence-informed forecasting. For demand-based event types whose models validate, and where authorised, the framework runs by month 48 as a **daily observation-mode dashboard** with Geneva's emergency services, forecasts and validity checks refreshed automatically, recorded and scored prospectively against observed outcomes, never yet steering care: the nucleus of a future living surveillance platform.

The project rests on an unusual combination: fifteen years of quantitative finance on rare-event risk and regime models, LiteRev-Evidence, and Swiss emergency data. Ambizione turns these into an independent research programme, led by me.

2. Proposed research

2.1 Current state of research in the field

2.1.1 Forecasting health-system crises: strong in steady state, weak at onset

Syndromic surveillance detects departures from expected baselines [Farrington 1996; Noufaily 2013], and forecast ensembles perform strongly in data-rich settings [Cramer 2022; Sherratt 2023]. Dispatch records lead confirmed surveillance: the 2009 A(H1N1) autumn onset was identified eight days early in three European regions [Rosenkötter 2013], long call series track influenza-like illness [Bonora 2025], and our systematic review maps the field [Edjinedja 2026].

The decisive limitation is **history dependence**: data-adaptive models need enough local observations to learn seasonality, weather response and crisis dynamics. Reviews of emergency-demand forecasting catalogue which model families perform for which outcome and horizon [Wargon 2009], and every recommended model is trained on years of local history.

The cold-start problem is a labelled-outcome problem. At crisis onset, context variables abound but the outcome to be forecast does not, and abundance itself can mislead: Google Flu Trends overestimated influenza by more than a factor of two [Lazer 2014]. Early signals do not close the gap: wastewater needs a shedding-to-incidence relationship that itself comes from external evidence, and transfer learning needs comparable contemporaneous observations. The common issue is **how to use external quantitative information without pretending it is perfectly transferable**.

2.1.2 The unused resource: published evidence as quantitative prior information

Bayesian borrowing is well developed: power priors [Ibrahim 2000], commensurate priors [Hobbs 2011], meta-analytic-predictive priors with robust conflict protection [Schmidli 2014]. Informative priors have been used to forecast disease under sparse local data [Cook 2023]. What has not been established is narrower and harder: whether an **automatically constructed** evidence prior, carrying its extraction error and its transportability uncertainty, improves **operational cold-start forecasts under strict historical information constraints**, and whether harmful borrowing is detected in time to act.

Parameter values are not the only transferable object. The literature also carries **explanatory variables and lag structures**, **model-form evidence** (which algorithm families work for which crisis type and horizon) and **outcome and threshold definitions**. Living systematic reviews keep syntheses current [Elliott 2014], yet to my knowledge no operational forecasting framework consumes these objects systematically.

Extraction. Automated extraction makes large-scale synthesis feasible, but numerical extraction remains less reliable than categorical (reported accuracy roughly 47–88% versus 74–96%), and omissions dominate the errors [Shankar 2026]. The open question is whether extraction and pooling preserve the dispersion a calibrated prior needs.

Transportability. Even perfectly extracted estimates may not transfer across populations, case definitions, health systems or policy regimes. Formal transportability tools exist [Bareinboim 2016; Dahabreh 2020; Degtiar 2023] but are not integrated into any operational cold-start framework.

2.1.3 Representing escalation as a state, not only a point forecast

Emergency operations often need an interpretable escalation state (routine, elevated, strained, critical) more than tomorrow's count. Latent-state representations are established in surveillance: Poisson hidden Markov models distinguish epidemic from non-epidemic periods [Le Strat 1999; Watkins 2009]; extreme-value methods frame rare critical exceedances [Coles 2001], applied to Swiss

hospital congestion [Ranjbar 2022]. Here they serve one purpose: a latent ordinal state as the **common representation for testing evidence borrowing at crisis onset**. Mechanistic transmission models [Keeling 2008] enter as components only where the crisis is epidemiological; environmental crises such as heat are driven by exposure–response, not transmission. Critical-slowing-down theory (rising variance and lag-1 autocorrelation before some transitions [Scheffer 2009]; epidemic applications [O'Regan 2013; Brett 2018; Southall 2021]) reads the shape of a short recent series: a **secondary information channel**, tested empirically.

2.1.4 From forecast accuracy to decision value

Proper scoring rules reward calibration [Gneiting 2007], but decision-analytic methods evaluate predictions under explicit consequences [Vickers 2006]: a forecast is operationally useful only when its probabilistic gain changes a defensible decision. Mortality-calibrated heat thresholds need not match demand thresholds [Lung 2021]; pollution–asthma emergency associations are heterogeneous across settings [Zheng 2015]. Methods are therefore evaluated first statistically, then under an elicited operational loss structure.

2.1.5 The specific gap addressed by this project

Ongoing programmes occupy the neighbouring ground: multi-model forecasting hubs run routinely (ECDC RespiCast, US CDC FluSight), Horizon Europe funds epidemic-intelligence infrastructure (including GeoAI4EI, in which I take part), and the Franco-Swiss GESICA programme builds cross-border crisis intelligence. Their stated objectives do not address whether automatically synthesised external quantitative evidence should enter a forecast as formal prior information at local cold start.

What nobody has established is whether these pieces connect around the **cold-start question**, and whether the loop from living evidence synthesis to variable choice, model choice, thresholds and automatically re-validated forecasts can be closed under pre-registered rules.

2.2 Current state of personal research and competences required for the project

My route is unusual, and it is why this project is tractable: fifteen years in quantitative finance (risk, extreme values, regime models, stress testing), then a doctorate in biomedical sciences at Geneva (defended 18 December 2023).

2.2.1 Prediction under sparse and imperfect information

In [Orel 2022] I predicted individual HIV status from socio-behavioural characteristics across East and Southern Africa, establishing where models transported between countries; [Merzouki 2021] and [Esra 2023] developed the theme. I now **senior-author** that line (last author, [Ng'ambi 2026], accepted: cardiovascular-disease classification across harmonised WHO STEPS surveys): I met this proposal's transportability problem first elsewhere, and my role on it has moved from conducting to directing.

2.2.2 Automated evidence synthesis

In [Orel 2023] I introduced **LiteRev**, an automated literature-review tool: natural-language or Boolean query across open databases, deduplication, corpus mapping, iterative relevance suggestion. I led its development with **Aziza Merzouki** (PhD, computer science) and raised its funding on my own initiative: **CHF 30,000** (UNIGE), **CHF 10,000** (Venture Kick), **CHF 20,000** (Mimosa), outside any group grant. The AI-in-EMS systematic review [Edjinedja 2026], within GESICA, uses it in its methods.

2.2.3 Outbreak and health-system modelling in Switzerland

In [Orel 2024] I compared clinical severity between Delta and Omicron sub-lineages in a Swiss tertiary centre. [Estill 2020] built age-structured scenario models for the Swiss SARS-CoV-2 epidemic under

planning time pressure: the experience this proposal's question comes from. I contributed to WHO African Region reporting and seroprevalence estimation [Nwosu 2021].

2.2.4 The prototype and test bed: LiteRev-Evidence

Since 2024 I have developed **LiteRev-Evidence**, extending LiteRev into structured quantitative extraction and modelling: the full review pipeline in working form, from a query searched live across the open literature, through deduplication, PRISMA-accounted screening and PICO extraction, to **provenance-tracked extraction graded for study-design certainty and pooled into parameter distributions**, and onward to the model step: candidate explanatory variables with their supporting articles; typed outcome definitions with ready-made emergency-demand targets (ED overload, bed occupancy, call surge); **model fitting across algorithm families with automated hyperparameter search, leakage-safe time-series validation and ranking on a stated metric**; and compartmental components, parameterised from extracted ranges, that can re-enter the statistical model as predictors. Per-scenario **living reviews re-run on schedule**, every update previewed before application: the evidence-to-model loop this proposal interrogates, in prototype form.

For GESICA I built the data foundation of the Geneva–Vaud–Neuchâtel network, whose Geneva core COLDSTART uses: **77 notifiable diseases in eight model classes**, and a referenced inventory of **28 surveillance sources (23 infectious, 5 environmental/non-infectious)** each documented for institution, coverage, resolution, latency, access and quality: a mapped, not assumed, data landscape; validation domains are chosen by **model class**.

The prototype makes the research feasible rather than aspirational: the fit-tune-rank mechanics T3.3's selection engine registers already run in embryonic form; the grant adds the science (two arms, registered rules, episode-level inference), not the plumbing, and the validated instrument is itself a deliverable.

2.2.5 Linked data on a contrasting crisis archetype

I lead the data work on a cantonal study **already under way** (BASEC 2026-00324, ethics granted) linking confirmed legionellosis cases in Geneva to individual domestic hot-water installations: a distinctive linkage, supplying a contrasting waterborne archetype.

Competences. Bayesian hierarchical and regime-switching estimation, extreme-value modelling, stress testing under misspecification, forecast evaluation and decision analysis come from the risk work and form the core of WP2–WP3; extraction and pooling from LiteRev-Evidence; domain knowledge from the doctorate, the Swiss COVID-19 work and GESICA.

2.3 Detailed research plan

2.3.1 Objectives and hypotheses

To determine **whether, and under what conditions, the chain from published quantitative evidence to operational forecasting closes at the onset of a health-system crisis: reliable extraction, transport to the local setting, improved probabilistic forecasts of escalation, early detection of harmful borrowing, decision-relevant gains, and an automated, prospectively verified loop.**

This end-to-end question decomposes into **one central hypothesis, H3a**; all else is subordinate, each row validating one link of the chain:

	Role	Statement
H1	Validation	Can the evidence be trusted enough to use?
C2	Adequacy criterion	Is the state representation fit to compare borrowing in?

	Role	Statement
H3a	**Central hypothesis**	**Does systematically extracted evidence improve cold-start forecast skill?**
H3b	Robustness	Is adaptive borrowing safe when the prior is wrong?
H3c	Secondary channel	Do resilience indicators add information beyond the evidence?
H4	Decision value	Is the gain large enough to change an operational choice?

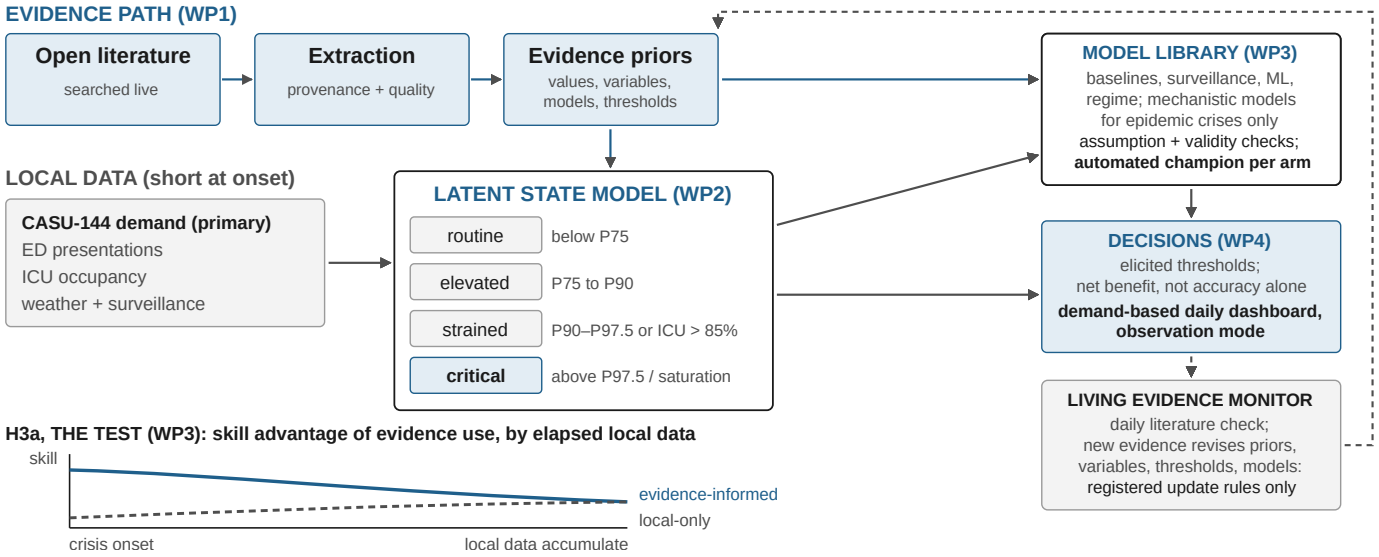


Figure 1: cold-start problem, evidence pipeline, model library with automated selection, living update loop.

O1: Make published evidence usable without hiding its uncertainty

H1. Automated extraction will systematically **understate the dispersion of the evidence-derived distribution** (reported within-study uncertainty and between-study heterogeneity), producing priors that are too concentrated; an explicit measurement-error layer will correct this under-dispersion, success assessed by a pre-specified criterion: coverage and calibration of intervals from the corrected distribution across held-out adjudicated studies.

Omissions dominate reported extraction errors [Shankar 2026], and whether they systematically shrink dispersion is exactly what H1 tests; if the loss cannot be corrected, the project establishes a boundary condition. The evidence target spans four object classes (T1.1); H1 is tested on the quantitative core, and classes 2 to 4 are validated externally against published syntheses (T1.2).

O2: Represent escalation in a form that separates state from the point forecast

C2, model adequacy criterion. The latent-state representation must yield **identifiable** parameters and **calibrated** escalation-state probabilities at matched false-alarm rates. Its role is the common state representation in which borrowing strategies are compared, not a claim that regime switching generally beats thresholding a point forecast.

T2.1's identifiability study and T3.3's calibration checks either establish adequacy or trigger the ordinal fallback; either outcome leaves H3a intact. Extreme-value modelling covers the critical tail; resilience indicators are **supporting covariates**.

O3: Test the cold-start hypothesis and map failure

H3a. An evidence-informed forecasting configuration (evidence-derived priors, explanatory-variable sets and model forms) improves probabilistic forecast skill during the early phase of a crisis; the persistence and decay of the advantage as local observations accumulate are assessed secondarily.

H3b. Adaptive borrowing that discounts the evidence when prior–data conflict emerges is **non-inferior** to fixed borrowing under well-specified priors, within a pre-specified margin Δ on the CRPS skill score, and is **superior** to fixed borrowing under deliberately misspecified priors.

Δ is fixed at one half of the minimal relevant H3a improvement, a pre-specified 50% preservation fraction (adaptive borrowing must retain at least half the gain that justifies borrowing at all): a maximum acceptable loss, not a simulation convenience.

H3c. Resilience indicators add predictive information beyond the evidence-derived prior and the local level/trend signal when the outcome history is short.

Primary confirmatory comparison (one, stated once): the **evidence-informed champion** against the **local-only champion**, both produced by the identical pre-registered automated procedure (T3.3), by **CRPS skill score**, over the pre-specified cold-start window, **on respiratory episodes only**. Heat repeats the identical contrast as a **sequential generalisation test**, run only if the respiratory test is met; the fixed order controls the family-wise error rate. The **shape of the advantage over elapsed local data** is reported: it should decay to nothing.

O4: Establish whether predictive improvement is decision-relevant

H4. For the demand-based archetypes, decision-analytic evaluation under the losses and escalation thresholds of emergency responders can rank modelling strategies differently from generic accuracy criteria; the evidence-derived strategy is useful only when its gain crosses a decision threshold.

Scope: three Geneva crisis archetypes, fixed for the grant

COLDSTART's real-world scope is three health-system crisis archetypes, and no more. (1) **Respiratory epidemics**: SARS-CoV-2, influenza and RSV, syndromically overlapping and surveilled together in Switzerland, carry the primary confirmatory test. (2) **Environmental heat events** carry the sequential generalisation test, with **air pollution (ozone, PM10) as co-exposure and effect modifier**, not a separate domain. (3) **Waterborne outbreaks**: Geneva legionellosis, the year-4 contrasting extension (T3.5). The domains are fixed for the project; further data serve benchmarking and sensitivity analyses only; no fourth operational domain is required. For respiratory and heat, both arms score **the same quantity, built the same way**: daily emergency demand from the CASU-144 series, restricted to the archetype's cause classes (T3.0): respiratory-related demand for epidemics; **heat-sensitive demand** for heat, Swiss evidence placing heat effects in dehydration, renal and psychiatric admissions, with a weak respiratory effect at older ages [Schulte 2024; Ragettli 2019]. Legionellosis instead forecasts case incidence (T3.5), outside this demand comparison. If the month-12 gate activates the registered fallback, both arms score it, built the same way, and the claim narrows.

What the project does not claim

The project does **not** aim to outperform forecast hubs in the data-rich regime, does not assume literature-derived priors are beneficial, and does not promise a clinically steering alarm by month 48: the dashboard runs strictly in observation mode. The contribution: **whether accumulated quantitative evidence can earn a formal role in forecasting before local outcome data become informative, and a map of when it should not be trusted.**

2.3.2 Work packages and methods

WP1: From published evidence to usable priors (M1–M20)

T1.1: Define the evidence target and register the protocol *(M1–M4)*

The evidence target spans four object classes; inclusion criteria, effect measures, uncertainty representation and transportability variables are fixed before extraction begins. **(1) Parameter**

values, the quantitative core (benchmarked in full): lag-structured **weather-demand coefficients** (daily mean temperature, heat-day exceedance, ozone, PM10); **surge magnitude and timing** (peak-to-baseline ratio, time to peak, onset growth rate); **length-of-stay and occupancy distributions** (ward and ICU); **admission fractions**; transmission parameters (R0/Rt, serial interval) and shedding-to-incidence conversions secondary. **(2) Explanatory variables and lag structures** reported as informative per crisis type. **(3) Model-form evidence:** which algorithm families the forecasting literature finds effective for which outcome, horizon and crisis type [Wargon 2009], feeding T3.3. **(4) Outcome and threshold definitions:** which series are used operationally and at what levels warnings fire [Lung 2021], entering T2.1's anchors and T4.1's elicitation as evidence-based starting points. Classes 2 to 4 carry provenance and certainty grades and are pursued only as inputs to the registered H3a configuration; H1's benchmark applies to class 1.

T1.2: Build the quantitative extraction benchmark *(M3–M9)*

I and a separately contracted independent extractor (§2.4) independently extract target quantities from a stratified random sample of **300** publications, with adjudication; stratification covers parameter class, reporting quality, study design and **source type** (journal versus timestamped situational report), so error is characterised separately for T3.2's faster secondary sources. An **open external layer** scores retrieval against the reported inclusion sets of published systematic reviews in the target classes, and extraction and pooling against their published summary estimates. The benchmark is an open deliverable.

T1.3: Characterise automated extraction error *(M7–M14)*

Compare automated extraction with the benchmark on point estimates, uncertainty, omissions and between-study dispersion; H1's primary metric is the median log-ratio of automated to adjudicated **standard deviation of the evidence-derived distribution**, per parameter class on a common scale under the registered pooling model, with a pre-specified interval (under-dispersion: ratio below one); within-study uncertainty and between-study heterogeneity are propagated separately and jointly, with sensitivity to the underlying model version.

T1.4: Construct uncertainty-aware evidence distributions *(M12–M18)*

Represent extraction error explicitly as measurement error; compare inverse-variance and quality-weighted pooling, meta-analytic-predictive priors and power-prior discounting. Output: a prior with provenance, extraction uncertainty and a documented borrowing weight.

T1.5: Transportability screen *(M15–M20)*

Characterise effect modifiers and study-setting differences relevant to Geneva; the transportability criteria, and the mapping from transportability score to prior widening or discounting, are fixed before the confirmatory evaluation, producing the metadata WP3's conflict analysis needs.

Deliverables. D1.1 open benchmark; D1.2 error analysis; D1.3 evidence-to-prior library.

Risk. If expert time binds, the benchmark narrows to the core parameter classes.

WP2: A parsimonious model of health-system escalation (M1–M28)

T2.1: Specify and identify the state model *(M1–M9)*

A Bayesian hierarchical Markov regime-switching model [Hamilton 1989] with an ordinal latent state (routine, elevated, strained, critical) observed through emergency calls (the primary channel), with ED presentations and ICU occupancy added where available; pre-specified weather, calendar and epidemic covariates; series-specific observation models sharing the state. Before real-data fitting: simulation-based identifiability and recovery, including the reduced-channel case; ordering constraints

resolve label switching; if separation is insufficient, the fallback is an ordinal state-space formulation; the criterion is state and transition recovery, not visual fit. **Provisional state anchors**, fixed at registration for WP2–WP3 (T4.1's later elicitation refines their decision interpretation, never the confirmatory state definition): **routine** below the 75th percentile of the seasonal baseline; **elevated** 75th to 90th; **strained** 90th to 97.5th, or sustained capacity pressure (ICU occupancy above 85%, ED boarding above its seasonal 90th percentile); **critical** above the 97.5th, or capacity saturation; the highest applicable state prevails, capacity saturation alone defining critical. Anchors are percentile-based per event type; in estimation they enter as priors on state-dependent levels, not hard cutoffs.

T2.2: Represent the critical tail *(M6–M14)*

Peaks-over-threshold/generalised Pareto modelling for rare exceedances, coupled to the critical-state probability, threshold sensitivity reported: supporting, not a separate objective.

T2.3: Introduce evidence-derived priors *(M10–M20)*

Map WP1 distributions to the parameters where published evidence is relevant (weather effects, surge magnitudes, transition characteristics); compare weakly informative, fixed evidence-derived and adaptive robust borrowing. The adaptive specification is a robust mixture of an evidence-derived and a weakly informative component; the conflict statistic and the discount function are fixed before the confirmatory evaluation, not tuned during it. Power-prior and commensurate-prior approaches are sensitivity comparators.

T2.4: Add resilience indicators as a secondary information channel *(M12–M20)*

Rolling variance and lag-1 autocorrelation, entered as optional covariates on transition dynamics: do they add information beyond level/trend and the evidence prior? A null result is interpretable.

T2.5: Calibration, automation and validity monitoring *(M20–M28)*

Calibration methods suited to temporal dependence; a conformal component [Angelopoulos 2023; Barber 2023] as a robustness layer if simulation confirms its assumptions. This task also builds the **automation layer**: public streams (weather, surveillance, wastewater) ingested automatically as they become available; models re-estimated at each input's native cadence with versioned re-fitting; and **assumption and validity diagnostics** on every fit (residual structure, dispersion, calibration): a model failing its checks is excluded by pre-registered rules. Release a reference implementation in LiteRev-Evidence.

Simulation is prior information, not data. Fitting literature-parameterised trajectories as if data would count the same information twice and disable the conflict diagnostic H3b depends on; simulation serves identifiability, structural constraints and prior predictive checks, never tightening the evidence prior.

Deliverables. D2.1 model specification and identifiability study; D2.2 open implementation; D2.3 methodological paper.

Risk. Weak regime separation. *Mitigation:* simulation first; the ordinal fallback; report the identifiability boundary as a result, not tuned until success.

WP3: The decisive cold-start experiment (M12–M42)

T3.0: Outcome hierarchy, data-access gate and episode eligibility *(M1–M14)*

Fixed before any evaluation is designed, never revisited on performance:

Outcome. The primary outcome is **daily respiratory-related emergency demand derived from CASU-144 records**, not a raw call count: a pre-registered classification of recorded call reasons and urgency levels. **The heat arm scores heat-sensitive demand**: the same series and construction,

restricted to cause classes fixed at registration (dehydration, renal, psychiatric), with respiratory-restricted and 75+ sensitivities. ED presentations and ICU occupancy enter as **additional observation channels on the shared latent state**; wastewater, sentinel consultations and weather as covariates.

Candidate outcome set and geography. The pre-registered candidates, in hierarchy order: (1) cause-filtered CASU-144 emergency demand (primary); (2) ED presentations by category; (3) ICU occupancy; (4) all-cause 144 engagements and hospital admissions as sensitivity series; open surveillance (Sentinella, wastewater) as fallback outcomes only. The unit is the **canton of Geneva at daily resolution** (the HUG centrale's coverage area): a deliberate single-canton design in the data ecosystem I already work in; generalisation comes from the contrasting archetypes and the open national series.

Data-access gate. CASU-144 is the pre-specified primary outcome. Criteria for **historical depth, resolution, latency and completeness** are fixed in advance; a month-12 gate confirms them; if not, the pre-specified fallback activates and the claim narrows.

Episode eligibility. An episode enters the confirmatory evaluation only if all of these are prospectively reconstructable: (1) a detectable onset under the prospective onset rule; (2) enough pre-onset history for its rolling baseline; (3) enough post-onset observations at the primary horizon; (4) the external evidence **as it stood at the historical origin**; (5) no leakage of future information; (6) separation from adjacent episodes. Failing episodes serve sensitivity analysis only. **At demand level, co-circulating pathogens form one episode.**

Two registration points. The hierarchy, gate criteria and eligibility rule are registered now. The window N, the archetype-specific horizons and the margin Δ are registered after the checkpoint and the episode inventory, before any evaluation runs, by T3.3's pre-declared selection rules.

Open-data validation track (M12–M24, pre-specified secondary). Before any clinical series connects, the complete rolling-origin machinery runs on open Swiss series requiring no agreement: federal respiratory surveillance, the national wastewater programme, weekly all-cause deaths, MeteoSwiss exposures. Archived multi-model hub forecasts give contemporaneous comparators for early, genuinely cold-start rounds [Cramer 2022; Sherratt 2023]. The track ships as a **public, re-runnable benchmark**: every component validated before operational data arrive, the methodological test robust to access outcomes; operational series sharpen the claim to emergency-system demand.

T3.1: Assemble the retrospective information set *(M12–M20)*

Harmonise the CASU-144 series with the additional channels and covariates; quantify completeness and delay; model right truncation and nowcasting [Höhle 2014; McGough 2020]; degraded reporting under strain could itself mimic an early-warning signal. Two Geneva channels extend the set: the HUG COVID-19 hospitalisation data I previously analysed [Orel 2024], re-requested as a secondary respiratory validation and calibration channel, and pharmacy sales with wastewater measurements via the pharmacien cantonal as syndromic covariates.

T3.2: Reconstruct the true information set *(M18–M30)*

For each historical onset, create successive forecast origins using **only information available at that date**. **Admissibility as a prior is defined by referent, not venue**: prior inputs describe *other* populations, places or past events. **The confirmatory prior uses peer-reviewed literature and preprints only**, under the rolling cut-off. Timestamped situational reporting on the ongoing event *elsewhere* (WHO/ECDC) enters as a **pre-specified secondary prior variant**, through the same extraction pipeline, outside the confirmatory contrast: H3a tests literature borrowing, not evidence

fusion. **Text describing the local event is excluded from any prior**: it is a noisy measurement of the outcome itself. Every input carries an index timestamp; revised sources enter as the **release vintage available at the origin**, never the final revised value, observation and release times both retained; origins re-run automatically through the T2.5 layer at each input's native cadence. Forecast horizons 7, 14 and 28 days (primary horizon per archetype fixed at the second registration point by a pre-specified operational criterion, never by forecast performance or episode results); the cold-start window counts local outcome observations **after a pre-defined real-time onset rule** that uses only variables available at the origin.

T3.3: The model library and the automated selection engine *(M20–M34)*

The library, registered before any evaluation, in two arms. Local-only: seasonal GLM with weekday and holiday terms; persistence; established surveillance exceedance and nowcasting (Farrington/Noufaily, Bayesian nowcasting [Salmon 2016]); penalised, gradient-boosted and upper-tail quantile learners on the pre-specified covariates; the WP2 regime model with weakly informative priors. **Evidence-informed:** the same candidates equipped with WP1 priors and design inputs, in three registered variants: fixed evidence-derived priors; adaptive borrowing with conflict monitoring; adaptive borrowing plus resilience indicators (H3c). **Mechanistic transmission components [Keeling 2008] are admissible only for epidemiological archetypes**; environmental archetypes use exposure–response structures; admissibility per archetype is registered.

Automated champion selection, nested against leakage. At each outer forecast origin, each arm's **champion** is chosen by a registered rule using only inner rolling validation on outcomes observed **before** that origin (CRPS the primary selection criterion, log score and calibration registered admissibility constraints, not ranking criteria; after T2.5 validity checks; blind to the other arm); the fixed champion then issues the outer forecast. Selection never sees the outcomes it is scored on. Identical selection machinery runs in both arms, so the primary contrast tests the full evidence-informed configuration, not a particular model family; the library is deliberately broad for selection, not hypothesis multiplication; registered secondary contrasts isolate the prior, variable-set and model-form components.

Confirmatory testing procedure (fixed-sequence).

1. **Test 1 (primary).** Evidence-informed champion vs local-only champion, CRPS skill score, **respiratory** episodes, cold-start window N, at $\alpha = 0.05$ two-sided (harm from borrowing is as decision-relevant as benefit). Passing requires paired-permutation $p < \alpha$ *and* a lower confidence bound above the pre-registered minimal relevant improvement.
2. **Test 2 (generalisation).** The identical contrast on **heat** episodes, scored on heat-sensitive demand (T3.0), at the same α, **conducted only if Test 1 passes**.
3. **If Test 1 fails**, H3a is not supported, Test 2 is exploratory, and the project's result is the failure map and the boundary condition.

All other contrasts are secondary. **The local-only arm is pinned in the registration**, its regime-model priors carrying a pre-declared band of vaguer and tighter alternatives as a sensitivity analysis: the comparator cannot become a straw man after the fact.

Primary endpoint: the CRPS skill score of the evidence-informed champion versus the local-only champion over the cold-start window. Aggregation is registered: per episode and arm, CRPS is averaged over eligible origins in the window at the primary horizon; the episode skill score is one minus the CRPS ratio (evidence-informed over local-only); the confirmatory statistic is its equally weighted mean over episodes. The unit of inference is the **episode**; eligible episodes are few (13–14

respiratory candidates before screening), so inference is by **paired permutation over episodes**, an **episode-level bootstrap** alongside (episodes are the resampling unit, kept intact, so within-episode temporal dependence is preserved), both pre-specified, disagreement reported. **The design is powered by simulation-based operating characteristics at episode level** (type I error, power, interval width under realistic dependence), run before the second registration point. The same simulation fixes, in advance, the **minimum number of eligible episodes** below which no confirmatory claim is made and the primary analysis becomes estimation, and the **selection rule** for N: the smallest grid value meeting the operating-characteristic criterion, chosen blind to forecast performance. The **minimal relevant improvement** is a registered value of the episode-level CRPS skill score, fixed before the second registration point by a pre-specified operational-relevance criterion defined before any outcome evaluation; T4.1's later elicitation translates observed improvements into operational consequences, it does not set the threshold. **Δ is registered at one half of this fixed value** (§2.3.1); the same simulation evaluates power for these targets and the stability of the ratio-based score under heterogeneous baseline CRPS. Secondary endpoints: log score, calibration (PIT, coverage), escalation detection at matched false-alarm rates. **H3b's Δ is fixed here, before any evaluation**: non-inferiority is a CRPS deficit no greater than Δ; superiority is tested on T3.4's misspecified priors. The well-specified case is defined in registered simulations whose evidence- and data-generating distributions match within a set tolerance; real episodes are complementary.

T3.4: Map benefit and failure *(M28–M38)*

Identify episodes where borrowing improves or worsens forecasts; characterise failure by population, outcome-definition, system, policy and temporal mismatch, extraction uncertainty and prior–data conflict, with deliberately misspecified priors (registered location, scale and transport shifts) as a stress test. The safety question: can adaptive discounting catch harmful borrowing in time? The output is a **failure map**, not an average.

T3.5: Test generalisation *(M34–M42)*

The sequential heat test and, resources permitting, the legionellosis extension (year 4): the borrowing framework transferred to forecasting case incidence at outbreak onset (low-count, case-based, on the linked case-environment series).

Deliverables. D3.1 reproducible cold-start evaluation pipeline; D3.2 primary result; D3.3 failure/stress-test map; D3.4 cross-archetype analysis.

Risk: operational data access. *Mitigation:* agreements initiated pre-award, letters accompany the application; and the open-data track already carries the methodological test under the same rolling-origin restriction, so delayed access narrows the claim (§2.4) without stalling the project.

WP4: From predictive skill to operational value (M24–M48)

T4.1: Elicit operational losses and thresholds *(M24–M32)*

Structured **SHELF** elicitation (n ≈ 15–20 across HUG emergency medicine, CASU-144 regulation and capacity management) elicits the consequences of early, late and unnecessary escalation and derives thresholds from the losses. **The primary decision is fixed in advance**: trigger surge-capacity escalation when the forecast probability of the strained/critical state crosses the elicited threshold. Non-state candidates map forecasts to escalation-state probabilities via T2.1's registered anchors, so every library member yields the same decision quantity.

T4.2: Decision-analytic evaluation and equity audit *(M30–M40)*

Re-evaluate the WP3 forecasts with net-benefit/decision-curve analysis and value-of-information: do rankings change once consequences are incorporated? Because operational records may encode structural differences, assess calibration and threshold performance across aggregate strata (age, sex, neighbourhood deprivation): calibrated on average but miscalibrated for a relevant group is not operationally ready.

T4.3: Counterfactuals, observation-mode dashboard and living updating *(M34–M48)*

Estimate, for selected episodes, what would have changed had escalation followed the model's signal (a simple capacity model, propagated uncertainty: counterfactuals, not causal estimates). For each archetype that passes T3.3 validation and T4.2's decision-value check, and subject to authorisation, the framework runs as a **daily observation-mode dashboard** with the partner services (144 regulation, ED, ICU): state probabilities, forecasts and validity checks refreshed automatically, **recorded and scored prospectively against the outcomes as they are observed, never used clinically**: the project ends with the system running and its quantification verified. The legionellosis extension is methodological and retrospective: it need not enter the capacity decision or the dashboard. A **living-evidence monitor** checks the literature daily and re-runs the per-scenario syntheses on a registered schedule [Elliott 2014], flagging new publications that would change a parameter value, a variable set, a threshold or the recommended model; changes apply only through version-controlled, pre-registered update rules, never silently. If authorisation is not granted, the project is complete on retrospective evaluation and says so.

Deliverables. D4.1 elicited loss structure and equity audit; D4.2 decision-analytic evaluation; D4.3 counterfactual analysis, with the observation-mode dashboard and living-update protocol where authorised.

Methods, data protection and reproducibility. Version-controlled R/Python and a registered analysis plan; clinical data processed in the UNIGE secure environment under the applicable institutional and regulatory approvals; benchmark and software released openly, synthetic equivalents where possible.

Expected outputs. Four to six papers and the durable open resources of §2.5; **I lead the methodological, benchmark and integrative outputs.** None of the explicit fallbacks above converts an inconclusive analysis into a success claim.

2.3.3 Research environment, team and resources

Host: Institute of Global Health, Faculty of Medicine, University of Geneva, with a methodological collaboration with **Data Science for Digital Health** (Prof. Douglas Teodoro) and a clinical one with **HUG emergency medicine** (Prof. Thibaut Desmettre; Dr Robert Larribau, CASU-144).

Institute of Global Health: epidemiology, infectious-disease modelling and automated evidence extraction in one institute: where the gap became visible. Independence is organisational: an independent programme alongside the Institute's groups (§2.6). **DS4DH** contributes the biomedical NLP WP1 needs; through GESICA I already work with the emergency-medicine partners; the AI-in-EMS review [Edjinedja 2026] evidences it.

Existing infrastructure. LiteRev-Evidence is a working prototype of the full evidence-to-model pipeline (§2.2.4); WP1 and WP2 turn it into a validated instrument and test it. The legionellosis study (BASEC 2026-00324, ethics granted) supplies the contrasting archetype. **Computing:** UNIGE HPC and the University's secure environment for clinical data.

Commitments, by status.

Item	Status	Evidence / action
Legionellosis linked data	**Secured**	BASEC 2026-00324; data via the médecin cantonal, SIG technical data
Host institute (ISG)	**Agreed**	Hosting accepted; SNSF-template signatures in preparation
Computing	**Secured**	UNIGE HPC and secure-environment access in place
DS4DH methodological collaboration	**Secured**	Letter of support signed
SNSF institutional confirmation letter	Requested	UNIGE Vice-Rectorate, standard administrative step
CASU-144 records, primary outcome	**Access held (GESICA)**	Continuous, daily, near real-time; project agreement and letter in preparation
ED presentations + ICU occupancy	In progress	Extraction in discussion with HUG; letters requested
COVID-19 hospitalisation series (HUG, daily)	Re-request planned	Previously analysed [Orel 2024]; validation and calibration series
Pharmacien cantonal: wastewater, pharmacy sales	Contacts established	Extracts to formalise, partly within GESICA
Open public series (surveillance, wastewater, deaths, weather)	**Public**	Freely accessible; no agreement needed

Remaining feasibility items: the ED/ICU extraction and the project 144 agreement; every row rests on an existing relationship.

Resources requested. Project funds are capped at **CHF 250,000 over four years**; the applicant's salary is covered separately. Doctoral students and postdocs cannot be employed under the 2026 rules; support personnel can.

Item	Rationale
Scientific/technical collaborator, **50% over 48 months**	T1.2 protocol, logistics and adjudication support; WP3 harmonisation and evaluation pipeline; reproducibility
Second independent extractor	Contracted M3–M9; required for T1.2's dual-extraction design
Data access, secure storage and processing	Project-specific extractions, secure-environment allocations and runs beyond existing infrastructure
Travel, conferences, incoming visits	Results presentation; international collaboration
Other eligible direct costs	As justified in the final SNSF budget

Preparatory work, before month 1 and outside grant funding: project-specific data agreements with HUG and 144/CASU (re-scoping access already held); **project-specific data governance and reuse**, confirmed under the applicable institutional and regulatory framework before any analysis; **infrastructure hardening** of LiteRev-Evidence before clinical data connect. Feasibility rests on **the prototype already existing** and on the fallbacks, each leaving the central question answerable.

2.4 Schedule and milestones

Who does what. I execute WP2 and WP3's confirmatory design personally and lead WP1's protocol and WP4's decision work. One **scientific/technical collaborator** (support personnel, 50% over 48

months) carries the T3.1–T3.2 harmonisation and rolling-origin pipeline and the reproducibility engineering behind D1.1 and D3.1; a separately contracted **independent extractor** (M3–M9) provides T1.2's second extraction arm. No other personnel are requested.

No serial chain: WP2 falls back to weakly informative priors if WP1 finds extraction inadequate; WP3's open-data track runs regardless of access; WP4's decision analysis is retrospective. **What open data alone costs:** open series are not emergency-system demand, so the claim narrows and most of WP4 turns illustrative; hence the agreements are a pre-award action.

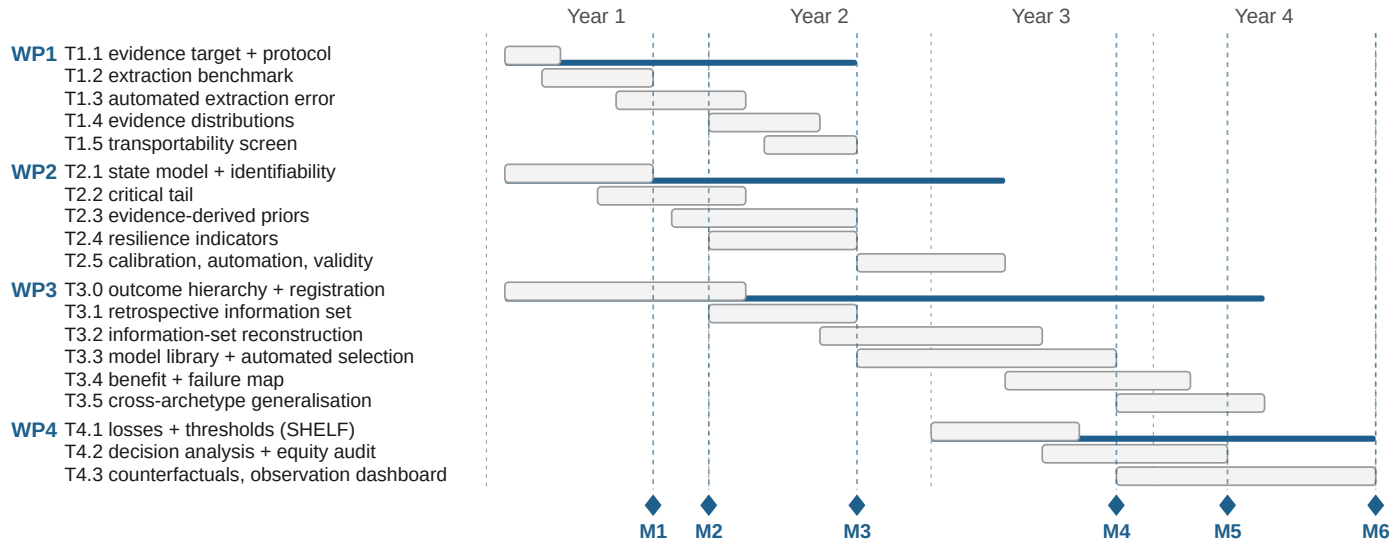


Figure 2: work packages, tasks and milestones over 48 months.

Milestone	Month	Criterion
M1	9	Extraction benchmark released
M2	12	State model passes identifiability criteria, or fallback selected
M3	20	Open-data benchmark running; operational series harmonised, or fallback activated
M4	34	Primary cold-start hypothesis tested by the champion contrast
M5	40	Decision relevance established
M6	48	Cross-archetype analysis complete; dashboard running and prospectively scored for validated demand-based archetypes where authorised

2.5 Relevance and impact

A null result changes identifiable practice. Modellers seed early-crisis models with parameters lifted from a few studies; evidence-synthesis platforms (mine included) assume pooled estimates are reusable as priors; a calibrated null tells both that the practice buys nothing where it matters most.

Form of publication. Peer-reviewed articles, preprints on submission, open access throughout; benchmarks, prior library and code as **citable, versioned open resources**; results returned to the partners. Durable resources, subordinate to the central question: **the extraction and replication benchmark**, **the public cold-start benchmark**, an **open reference framework** and a **decision-analytic evaluation framework**.

Practical and societal impact. In a crisis's first weeks, emergency systems decide whether to open capacity, redistribute resources or escalate before local outcomes are informative; the project quantifies whether external evidence improves those decisions, and with what residual uncertainty.

Where this points operationally. Each validated demand-based archetype leaves a registered, automatically re-validated configuration and, where authorised, a daily observation-mode dashboard, running and prospectively verified, in the hands of the services that would use it. Extending that nucleus into a multi-crisis surveillance platform is the explicit post-grant pathway; this project supplies the validated core and the update rules.

2.6 Relevance for personal career development

Why I am ready, and why now. I have the methods, the prototype and the domain access, used so far inside other people's programmes; what I lacked is a research agenda of my own.

Programme	Objective	My role	Distinction from Ambizione
GESICA	Franco-Swiss crisis intelligence	Contributor	Decision support; no test of evidence-derived priors
GeoAI4EI (Horizon Europe)	European epidemic-intelligence toolbox	Contributor	Infrastructure; no test of borrowed evidence
Legionellosis study (ongoing)	Geneva waterborne outbreak determinants	Data lead	Supplies a contrasting validation extension
Ambizione	**When evidence borrowing helps at cold start, and when it harms**	**PI, full-time**	-

This project does not build another epidemic-intelligence platform: it uses existing infrastructures to test a question that is a work package of none of them.

The question and the methods are mine. The question arose while I was building LiteRev-Evidence: the platform could pool published estimates into priors, and nothing established whether they improved forecasting in a new setting. The methodological core came from fifteen years of quantitative finance, from no group I have worked in; the platform was developed at UNIGE under my scientific direction.

Collaborators and their roles. Prof. Olivia Keiser leads the group in which I developed LiteRev; she also leads the legionellosis study, GeoAI4EI and the MAS on which I teach; she co-authors my output (expected at this stage) and has **no scientific role in this project**. Prof. Douglas Teodoro leads GESICA and provides the WP1 biomedical-NLP collaboration.

The organisational position is the transition. Hosted by the **Institute of Global Health**, I will lead an independent programme alongside its groups rather than inside one; the guarantees are in the confirmation letters. The programme is executable by the PI and budgeted support personnel: **no named individual other than the PI is scientifically load-bearing.**

Where this leads. By month 48: a validated answer and a first-author record: the basis for group leadership or a professorship. Ambizione consolidates my research time into one programme.

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